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// Name: Utkarsha Patil
// Roll no: SEA52
// Dynamic Text Editor Using Singly Linked List

#include<stdio.h>
#include <stdlib.h>
typedef struct SLL{
   char data[10];
   struct SLL *next;
}SLL;
void main(){
   SLL *head, *p, *q, *r, *nw, *temp;
   head = NULL;
   int ans, op, loc;
   do{
      printf("\n1. Create\n2. Insert\n3. Display\n4. Delete\n5. Reverse\n6. Exit\n");
      printf("Enter option: ");
      scanf("%d", &op);
      switch(op){
         case 1:
         do{
            nw = (SLL*)malloc(sizeof(SLL));
            printf("Enter data: ");
            scanf("%s", &nw->data);
            nw -> next = NULL;
            if(head == NULL){
               p = head = nw;
            }else{
               p ->next = nw;
               p = nw;
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   }
   printf("Do you want to continue: ");
   scanf("%d", &ans);
}while(ans==1);
break;
case 2:
printf("Enter location: ");
scanf("%d", &loc);
p = (SLL*)malloc(sizeof(SLL));
printf("Enter data: ");
scanf("%s", &p->data);
if(loc == 1){
   p -> next = head;
   head = p;
}else{
   q = head;
   for(int i = 1; i < loc-1; i++){
      q = q -> next;
      if(q == NULL){
         printf("underflow");
      }   }
   p -> next = q -> next;
   q -> next = p;
}
break;
case 3:
for(p = head; p != NULL; p = p -> next){
   if(p == NULL){
      printf("EMPTY");
   }
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   printf("%s->", p -> data);
}
printf("NULL\n");
break;
case 4:
printf("Enter location: ");
scanf("%d", &loc);
if(loc == 1){
   head = head -> next;
   free(p);
}else{
   q = head;
   for(int i = 1; i < loc-1; i++){
      q = q -> next;
      if(q == NULL){
         printf("underflow");
      } }
   p = q -> next;
   q -> next = p -> next;
   free(p);
}
break;
case 5:
p = NULL; //previous node
q = head; //current node
r = NULL; // next node
while(q != NULL){
   r = q -> next;
   q -> next = p;
   p = q;
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            q = r;
        }
        head = p;
        for(temp = head; temp != NULL; temp = temp -> next){
            if(temp == NULL){
                printf("underflow");
            }
            printf("%s->", temp -> data);
        }
        printf("NULL\n");
        break;
        case 6:
        printf("EXITED SUCCESSFULLY");
        break;
        default:
        printf("Invalid option");
      }
   }while(op != 6);
}

// Output
1. Create
2. Insert
3. Display
4. Delete
5. Reverse
6. Exit
Enter option: 1
Enter data: vaishnavi
Do you want to continue: 1
```

Enter data: arya

Do you want to continue: 1

Enter data: purva

Do you want to continue: 1

Enter data: saee

Do you want to continue: 0

1. Create

2. Insert

3. Display

4. Delete

5. Reverse

6. Exit

Enter option: 3

vaishnavi->arya->purva->saee->NULL

1. Create

2. Insert

3. Display

4. Delete

5. Reverse

6. Exit

Enter option: 2

Enter location: 3

Enter data: shreya

1. Create

2. Insert

3. Display

4. Delete

5. Reverse

6. Exit

Enter option: 3

vaishnavi->arya->shreya->purva->saee->NULL

1. Create

2. Insert

3. Display

4. Delete

5. Reverse

6. Exit

Enter option: 4

Enter location: 5

1. Create

2. Insert

3. Display

4. Delete

5. Reverse

6. Exit

Enter option: 3

vaishnavi->arya->shreya->purva->NULL

1. Create

2. Insert

3. Display

4. Delete

5. Reverse

6. Exit

Enter option: 5

purva->shreya->arya->vaishnavi->NULL

1. Create

2. Insert

3. Display

4. Delete

5. Reverse

6. Exit

Enter option: 6

EXITED SUCCESSFULLY